Glassy State Simulation using Lennard-Jones Potential

Project Report
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Abstract

This project simulates a two-dimensional system of interacting particles using the Lennard-Jones potential and minimizes the total potential energy using the Steepest Descent optimization algorithm. The objective is to obtain a stable low-energy (glassy) configuration through numerical optimization.

Project Objectives

• Simulate a 2D particle system.

• Model particle interactions using the Lennard-Jones potential.

• Compute forces acting on every particle.

• Minimize the total potential energy.

• Obtain the final glassy-state configuration.

Introduction

Understanding how particles interact and arrange themselves into stable configurations is fundamental in computational physics and materials science. Glassy states represent local minima in the potential energy landscape. This project demonstrates how numerical optimization can be used to relax a particle system from an initial configuration to a low-energy state.

Theory

The Lennard-Jones potential models both attractive and repulsive interactions between particles. Short-range repulsion prevents particles from overlapping, while long-range attraction stabilizes the system. The force acting on each particle is computed from the negative gradient of the potential energy. Particle positions are updated iteratively using the Steepest Descent algorithm until convergence.

Methodology

1. Generate an initial 2D particle lattice.

2. Compute pairwise Lennard-Jones interactions.

3. Calculate the force acting on each particle.

4. Update particle positions using Steepest Descent.

5. Recalculate energy and forces.

6. Repeat until convergence.

7. Save the optimized particle configuration and performance metrics.

Simulation Parameters

• Number of particles: 1024

• Dimensions: 2D

• Potential: Lennard-Jones

• Optimization: Steepest Descent

• Maximum iterations: 3000

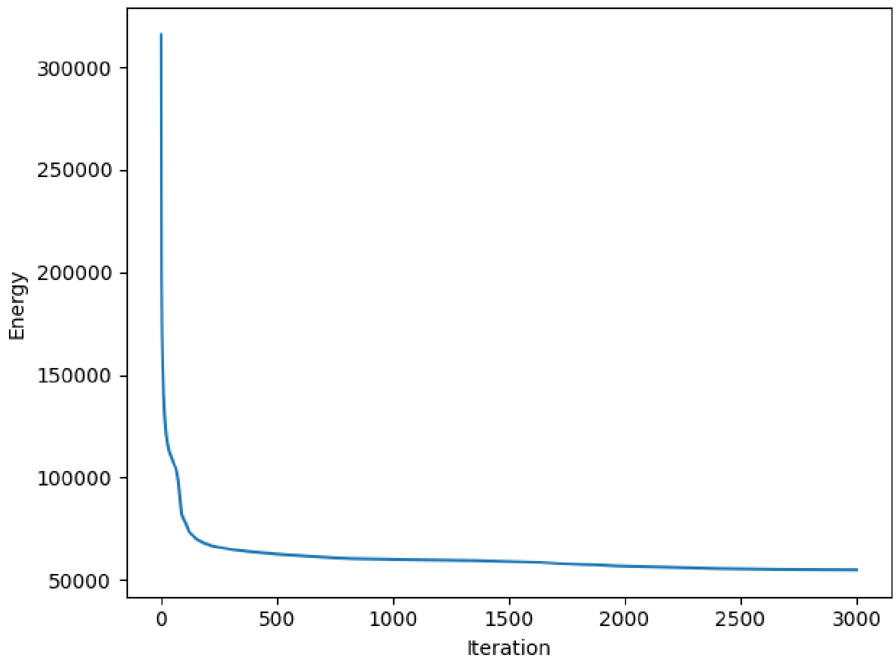
• Step size (α): 0.02

• $\varepsilon = 1.0$

• $\sigma = 1.0$

Results

The energy of the system decreases rapidly during the initial iterations and gradually converges to a stable value, indicating successful optimization. The maximum force acting on particles also decreases throughout the simulation, confirming convergence. Comparison of the initial and final particle configurations shows structural relaxation and particle displacement toward a low-energy glassy state.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250000
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0
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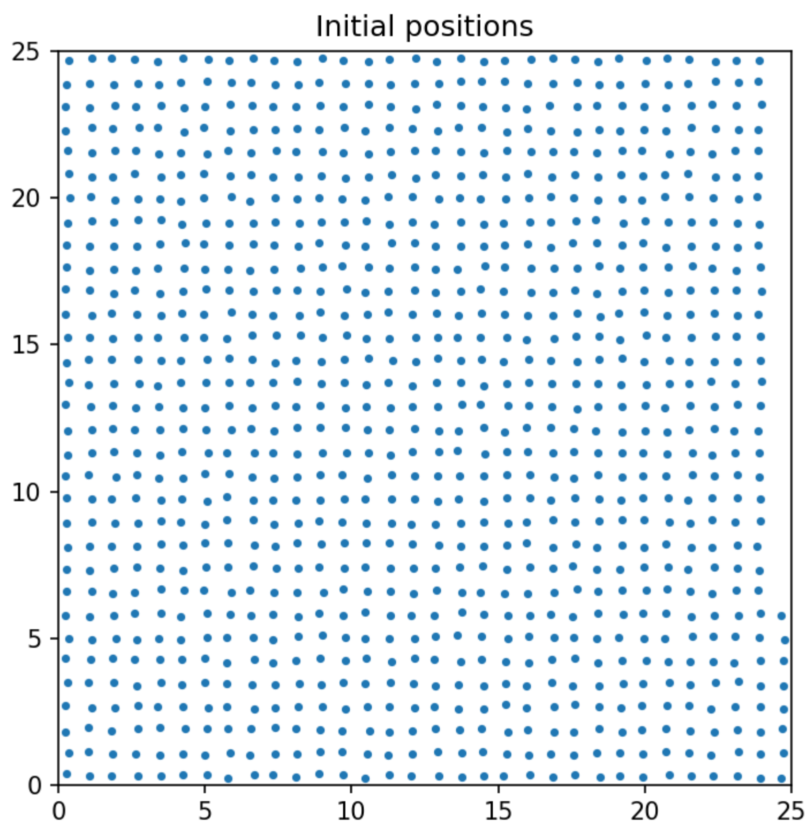
1000
1500
2000
2500
3000
Iteration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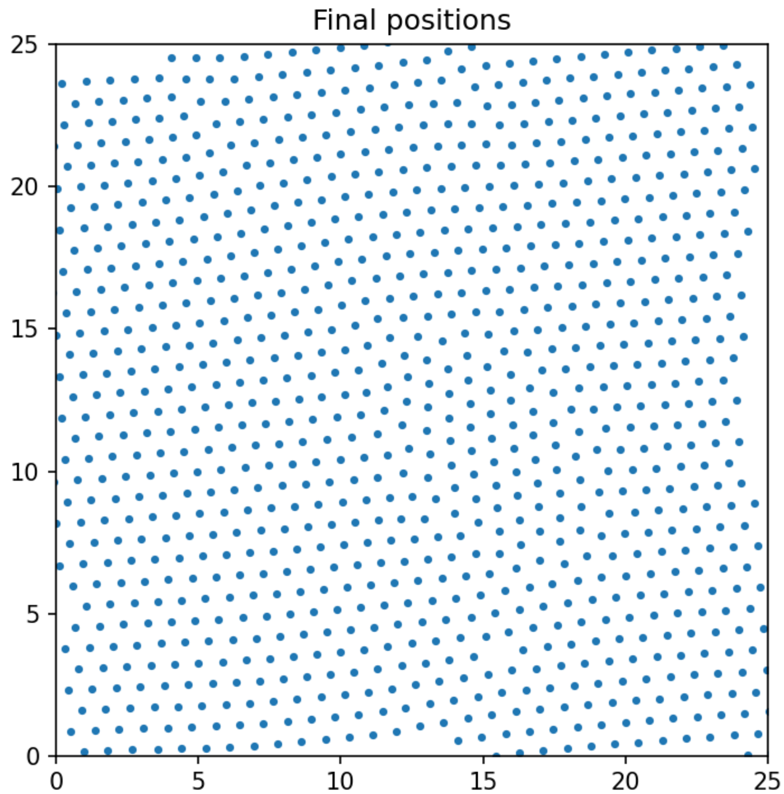
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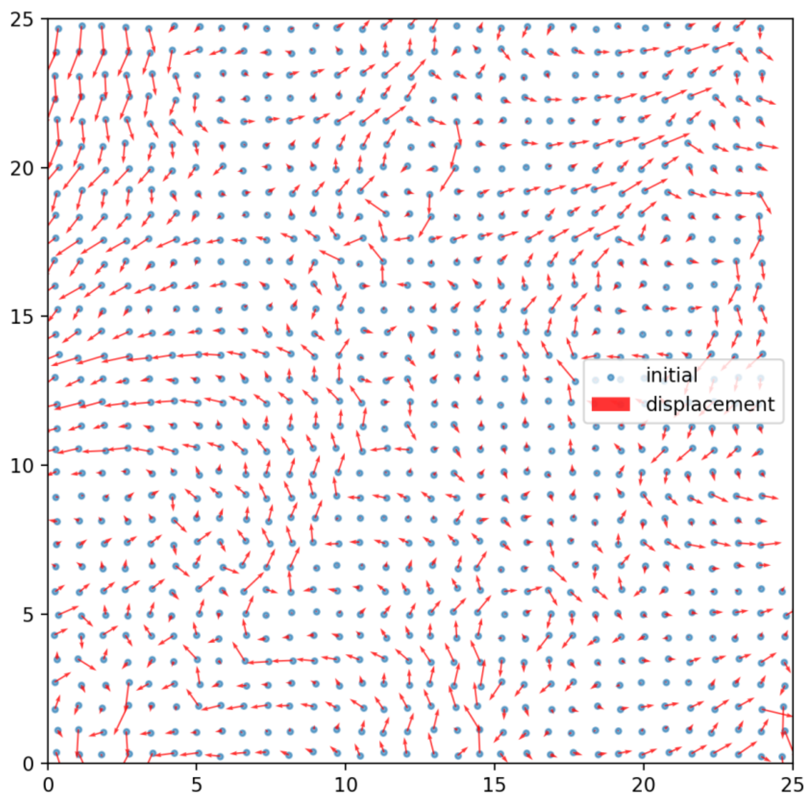
initial
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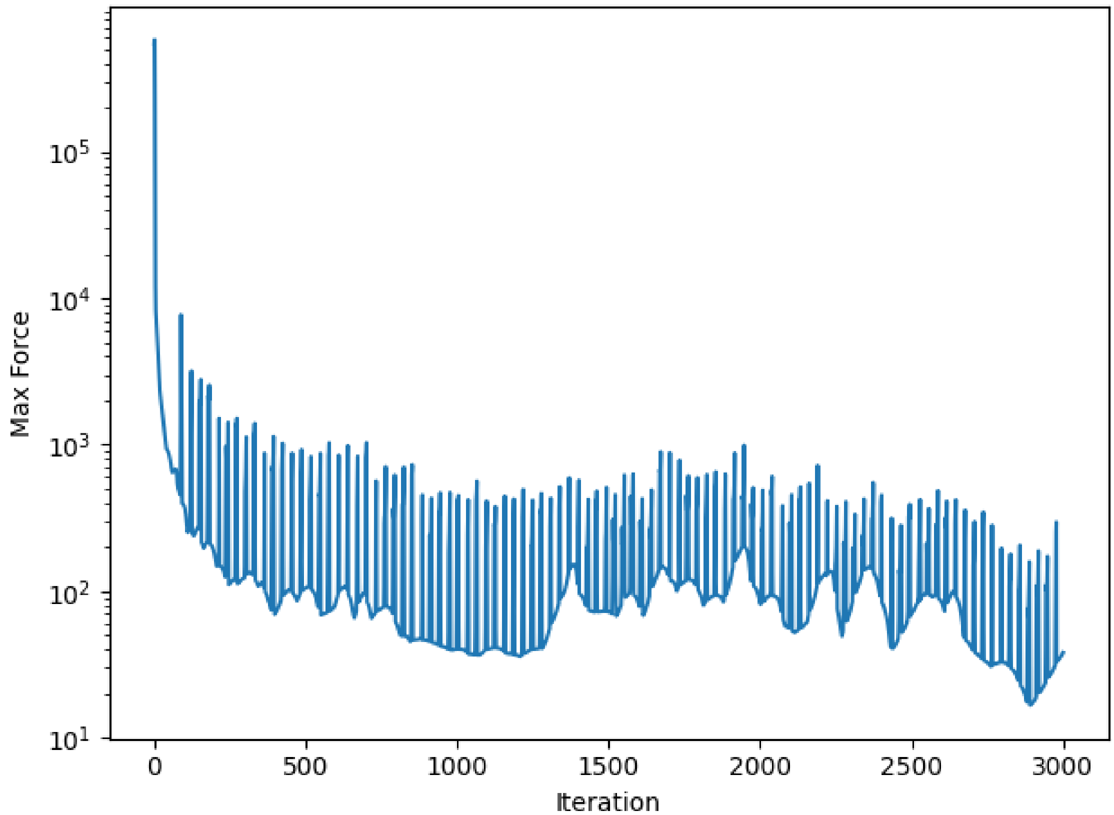

Challenges

• Selecting an appropriate learning rate.

• Maintaining numerical stability.

• Efficient computation of pairwise interactions.

• Ensuring convergence without oscillations.

Applications

• Molecular dynamics

• Materials science

• Glass formation studies

• Computational chemistry

• Scientific computing

Conclusion

The project successfully demonstrates numerical energy minimization using the Lennard-Jones potential and the Steepest Descent algorithm. Starting from an ordered particle arrangement, the simulation converges to a stable glassy-state configuration with significantly lower potential energy, illustrating the effectiveness of gradient-based optimization in computational physics.